

CSE 401: Numerical Analysis, Simulation and Modeling

Online 3: Monte Carlo Estimation of π (Buffon's Needle Problem)

1. Problem Statement

You are required to estimate the mathematical constant π using a **Monte Carlo Simulation** of the classical **Buffon's Needle Experiment**. A needle of length $L = 1.0$ is dropped randomly onto a flat surface ruled with parallel lines spaced at a uniform distance $D = 2.0$ apart ($L \leq D$).

Given System Parameters:

- Needle Length: $L = 1.0$
- Spacing between Parallel Lines: $D = 2.0$
- Condition: $L \leq D$ (Needle cannot cross more than one line simultaneously)

2. Mathematical Modeling & Random Variables

Each random drop of the needle is uniquely defined by **two independent continuous uniform random variables**:

1. Needle Center Position (X): The perpendicular distance from the midpoint (center) of the needle to the nearest parallel line.

- Distribution: $X \sim U(0, D/2) = U(0, 1.0)$
- Generation: Using uniform random number $R_1 \in [0, 1)$: $X = (D / 2) \times R_1 = 1.0 \times R_1$

2. Needle Orientation Angle (θ): The acute angle formed between the needle and the parallel lines.

- Distribution: $\theta \sim U(0, \pi/2)$
- Generation: Using uniform random number $R_2 \in [0, 1)$: $\theta = (\pi / 2) \times R_2$

3. Hit (Line Crossing) Condition

A needle crosses (hits) a line if and only if the center distance X is less than or equal to the perpendicular projection of half the needle length:

$$\text{Hit Condition: } X \leq (L / 2) \times \sin(\theta) \implies X \leq 0.5 \times \sin(\theta)$$

4. Theoretical Probability & π Estimator

The theoretical crossing probability P is calculated by integrating the hit region over the sample space:

$$P = \left[\int_0^{\pi/2} (L/2) \sin(\theta) d\theta \right] / \left[(D/2) (\pi/2) \right] = (2L) / (\pi D)$$

For $L = 1.0$ and $D = 2.0$, the exact crossing probability is: $P = 2(1) / (2\pi) = 1 / \pi \approx 0.31831$.

In a simulation of N total drops with H total hits (crossings):

- Empirical Crossing Probability: $P_{\text{est}} = H / N$
- Monte Carlo Estimator for π : $\pi_{\text{est}} = (2L) / (D \times P_{\text{est}}) = (2 \times L \times N) / (D \times H)$
- For $L = 1, D = 2$: $\pi_{\text{est}} = N / H = 1 / P_{\text{est}}$

5. Implementation Requirements

Write a Python program that performs the following:

1. Implements the simulation function `buffon_needle_pi(n_drops, L=1.0, D=2.0)`.
2. For each drop, generates $x = (D / 2.0) * \text{random.random}()$ and $\theta = (\text{math.pi} / 2.0) * \text{random.random}()$.
3. Tests condition $x \leq (L / 2.0) * \text{math.sin}(\theta)$ and counts total hits H .
4. Computes $p_hat = H / n_drops$ and $pi_estimate = (2.0 * L) / (D * p_hat)$.
5. Executes the simulation for $N \in \{100, 1000, 10000, 100000, 1000000\}$ and displays the formatted results.

6. Expected Sample Output

N (Drops)	Hits (H)	p_hat (H/N)	Pi Estimate	Absolute Error
100	32	0.32000	3.125000	0.016593
1,000	315	0.31500	3.174603	0.033011
10,000	3,184	0.31840	3.140703	0.000889
100,000	31,812	0.31812	3.143468	0.001875
1,000,000	318,310	0.31831	3.141592	0.000001